

Alex Short

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Education

California Polytechnic State University, San Luis Obispo (Since Fall 2021)		Estimated Completion:
Major:	Bachelor of Science in Physics	June 2025
2 nd Major:	Bachelor of Science in Computer Engineering	June 2025
Minor:	Mathematics	Completed June 2023

Objective & Interests

I am currently applying to graduate school in theoretical physics in order to gain more advanced knowledge to support my desired physics research and career. I want to become an academic, teach as a professor at a university, and conduct cutting edge theoretical physics research, focusing on quantum gravity, holographic dualities, and quantum field theory. Through my career in undergrad I have garnered a passion for working out physics problems, helping others grow academically, and developing presentations to teach complicated high level subjects to diverse audiences. I am looking to become a graduate student at a university to further these skills and grow these skills into a future career in physics.

Current & Past Research Projects

AdS/CFT Research | Date: Future | Advisor: Dr. Lei Lu

- I will explore the AdS/CFT dynamics of the magnetic black brane to learn about **holographic renormalization** and how quantum fields can be correlated to higher dimensional **gravitational systems**
- Learning **String Theory** and Large N QCD as a method of learning the gauge/gravity duality

Loop Quantum Gravity | Date: Present | Advisor: Dr. Lei Lu

- Exploring the canonical quantization of **general relativity** using **Yang-Mills** theories and **Chern-Simons** theories to develop a pedagogical publication on the subject of **loop quantum gravity**
- Use of Ashtekar's Variables to construct a canonical quantization of GR and write a pedagogical review of the entire field of Loop Quantum Gravity

SLAC Internship | Date: July - August 2024 | Advisor: Claudio Emma

- Interned with the SULI program at the **SLAC National Laboratory** at **Stanford University**
- Worked on simulations of XTCAV components for the FACET-II Accelerator

University of Colorado Boulder REU | Date: July – August 2023 |

Advisors: Dr. Joe MacLennan & Dr. Cheol Park

- Observation of the diffusive properties of smectic C islands in liquid crystals
- Use of polarized microscopy to **observe topological vortices** in liquid crystal films
- Use of artificial intelligence to track and quantify the diffusion of features in liquid crystal films, as well as image processing and statistical analysis of the results

Frost SURP: Shape Dynamics & Gravitational Physics | Date: February – August 2022 |

Advisor: Dr. Sean Echols & Dr. Julian Barbour

- Use of simulations to understand **classical gravitational** systems
- Modeling and understanding of ‘shape space’ in the classical sense in order to find **quantum properties of gravitation**

Diffusion in Microhydrodynamical Systems | Date: Present | Advisor: Dr. Jon Fernsler

- Determination the effects of fluid flow on microscopic diffusion present in quasi-two dimensional systems (fluid-thermal coupling in viscous fluids)
- Use of **fluorescence microscopy** to observe lipid monolayers in a Langmuir trough setup in order to quantify diffusion
- Exploration of the aggregative properties of quantum dots embedded in a thin smectic liquid crystal film

USB to Automotive-Ethernet Adapter for Zoox | Date: Present | Advisor: Dr. John Oliver

- I am the **Project Manager** of a small team designing a self-contained adapter from USB 3.0 to 1000 BaseT Automotive Ethernet
- Skills: PCB design, circuit design, leadership skills, project organization
- Our client is Zoox, a self driving car company

Past Presentations

→ **“Island Dynamics in Freely Suspended Thermotropic Liquid Crystal Films” | Place: CU Boulder Physics REU Conference | Date: August 2023**

- ◆ Scientific talk about the elastic and hydrodynamic coupling in anisotropic fluids, as well as my experimental results

→ **“Transverse Deflecting Cavities: Measuring the Particle Beam” | Place: SLAC National Lab Student Intern Presentations | Date: August 2024**

- ◆ Scientific talk about my work in analyzing and simulation XTCAV components in the FACET-II accelerator during my internship at SLAC

→ **“Exploring the Structure of Shape Space” | Place: FROST Research Conference 2022 | Date: August 2022**

- ◆ Scientific talk about my work in analyzing and simulation XTCAV components in the FACET-II accelerator during my internship at SLAC

→ **“Non-Equilibrium Fluctuations and Diffusion in quasi-2D Liquids” | Place: BCOSAM Spring Research Conference 2024 | Date: May 2024**

- ◆ Poster Presentation about hydrodynamics in two dimensional fluids, and current experiments

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→ **“USB - Automotive Ethernet Adapter” | Place: Computer Engineering Capstone Midpoint Presentations | Date: November 2024**

- ◆ Poster Presentation about hydrodynamics in two dimensional fluids, and current experiments
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Relevant Skills & Knowledge

- C, C++, Mathematica, Matlab, Python, Java, HTML/CSS, Verilog, Systemverilog, RISC-V Assembly, ARM Assembly, x86 Assembly
- Parallel Computing in Matlab & C/C++
- Mathematical Skills: Advanced Tensor Calculus, Field Theory, Real Analysis, Linear Algebra, Abstract Algebra, Partial Differential Equations, Differential Geometry
- FPGA Design, Implementation, and Programming
- Microcontroller Design, Implementation, and Programming (Including Arduino, Nano, and STM32)
- Multi-layer PCB design and layout
- Electronic circuit design and digital signal processing
- Computer graphics, raytracing, and graphics hardware (GPU architecture)
- Physics knowledge (outside of classes taken): String Theory, Loop Quantum Gravity, AdS/CFT, Quantum Field Theory

Relevant Coursework & Reading

List of courses taken in undergraduate studies relevant to my research interests:

- Phys418: General Relativity
- Phys403: Particle & Nuclear Physics
- Phys406: Quantum Mechanics II

List of books & review papers I have read which are relevant to my research interests:

- “Loops, Knots, Gauge Theories and Quantum Gravity” by R. Gambini and J. Pullin
- “A First Course in Loop Quantum Gravity” by R. Gambini and J. Pullin
- “Gauge Fields, Knots, and Gravity” by J. Baez and J. Muniain
- “The AdS/CFT Duality Guide” by M. Natsuume
- “Large N Field Theories, String Theory and Gravity” by O. Aharony, S. Gubser, J. Maldacena, H. Ooguri, and Y. Oz
- “Lie Algebras in Particle Physics” by H. Georgi
- “Spacetime and Geometry” by S. Carroll
- Selections from “Foundations of Differential Geometry” by Kobayashi & Nomizu
- “Classical Mechanics” by H. Goldstein, C. Poole, and J. Safko
- Chapters 1-20 from “A Standard Model Workbook” by T. Moore
- “Applied Partial Differential Equations” by R. Haberman
- “The Principles of Electromagnetic Theory and Relativity” by M. Tonnelat

